

Syu-Huei (Frank) Huang

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EDUCATION

University of Florida (UF)

Aug 2025 - May 2027

M.S. in Electrical & Computer Engineering

Gainesville, FL

- GPA: 4.0 / 4.0
- Achievement Award Scholarship
- Selected Coursework:* Applied Machine Learning, Autonomous Robotics, Medical Image Analysis

National Taiwan University of Science and Technology (NTUST)

Sep 2021 - Jun 2023

M.S. in Electrical Engineering

Taipei, Taiwan

Chung Yuan Christian University (CYCU)

Sep 2017 - Jun 2021

B.S. in Electrical Engineering

Taoyuan, Taiwan

TECHNICAL SKILLS

Languages: Python, C/C++, Bash, MATLAB

Machine Learning: PyTorch, TensorFlow, Deep Learning, Transformers, Representation Learning, Computer Vision, CNNs

Robotics & Autonomous Systems: ROS 2, CARLA, LiDAR SLAM, Sensor Fusion, Path Planning, Inverse Kinematics

Systems & Deployment: CUDA, NVIDIA Jetson Nano/TX2, Linux, Git, AWS EC2, Apache Spark, Apache Kafka, Docker

WORK EXPERIENCE

Gong Lab

May 2026 - Present

Research Assistant @ UF

Gainesville, FL

Advisor: Dr. Kuang Gong

- Research machine learning methods for multimodal medical image quality assessment using Multimodal Large Language Models (MLLMs).
- Develop PyTorch training and evaluation pipelines to benchmark MLLMs, investigating visual-language alignment, multimodal reasoning, and domain generalization.

IoT Smart Living Laboratory

Sep 2021 - Jun 2023

Research Assistant @ NTUST

Taipei, Taiwan

Advisor: Dr. Ching-Hu Lu

- Developed an end-to-end dense video captioning pipeline for edge cameras using PyTorch, improving BLEU-4 by **12.6%** and CIDEr by **22.5%** over prior baselines.
- Designed and optimized a lightweight video understanding model that reduced parameter count by **56.9%** while maintaining competitive inference accuracy.
- Optimized PyTorch training and inference workflows for edge AI applications.

PROJECTS

AutoEncoder-based Anomaly Detection

May 2026

- Developed a fully unsupervised video anomaly detection framework using a convolutional autoencoder with a novel latent entropy loss.
- Proposed a differentiable **Minimal Latent Entropy (MLE)** loss to improve latent representation learning and anomaly separability.
- Achieved up to **0.970 AUC** on public benchmarks, outperforming existing fully unsupervised baselines.

VLM-Guided Autonomous Mobile Manipulation System

Feb 2026

- Built an autonomous mobile manipulation pipeline integrating **ROS 2**, **OWLv2**, and **LiDAR SLAM** for navigation and object retrieval.
- Implemented **A*** path planning, **ROS 2** feedback control, and inverse-kinematics-based robotic arm manipulation for object grasping and placement tasks.

CT-MR Thorax-abdomen Intra-patient Registration based on Diffeomorphic VoxelMorph

Sep 2025

- Developed a Diffeomorphic VoxelMorph-based medical image registration framework that improved mutual information by **over 100%** and reduced folding artifacts to **0.08%** for CT-MR alignment.

PUBLICATIONS

[1] **S.-H. Huang**, C.-H. Lu, "Sequence-Aware Learnable Sparse Mask for Frame-Selectable End-to-End Dense Video Captioning for IoT Smart Cameras," *IEEE IoT Journal*, 2023. [\[paper\]](#)

[2] G.-T. Liu, C.-H. Lu, **S.-H. Huang**, "Hand-detection with Transferrable Design for Smart Factories," *ISPACS*, 2021. [\[paper\]](#)